Chapter 09 Part 2: Rate Law

1.

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

A sample of N_2O_5 was placed in an evacuated container, and the reaction represented above occurred. The value of $P_{N_2O_5}$, the partial pressure of $N_2O_5(g)$, was measured during the reaction and recorded in the table below.

Time (min)	$P_{N_2O_5}$	In($P_{N_2O_5}$)	$\frac{1}{P_{N_2O_5}}$ (atm^{-1})
0	150	5.0	0.0067
100	75	4.3	0.013
200	38	3.6	0.027
300	19	2.9	0.053

Which of the following correctly describes the reaction?

(A) The decomposition of N_2O_5 is a zero-order reaction.

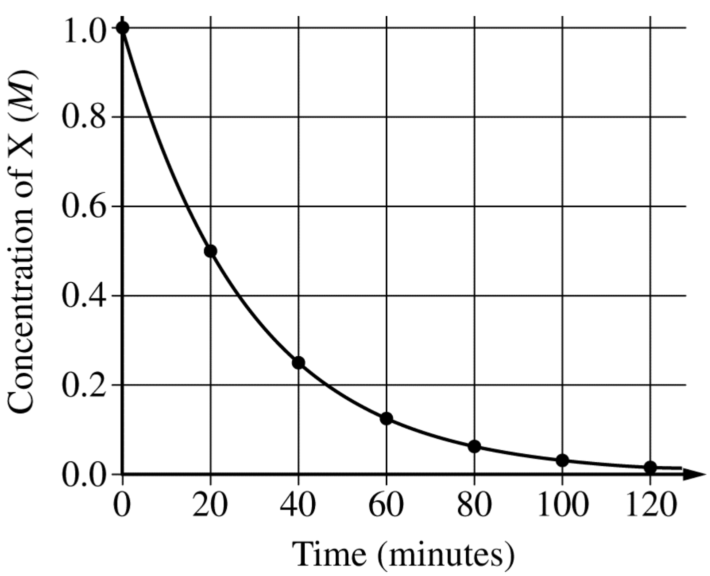
(B) The decomposition of N_2O_5 is a first-order reaction.

(C) The decomposition of N_2O_5 is a second-order reaction.

(D) The overall reaction order is 3.

2. X → Products

A student studied the kinetics of the reaction represented above by measuring the concentration of the reactant, X, over time. The data are plotted in the graph below.



Which of the following procedures will allow the student to determine the rate constant, k , for the reaction?

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(A) Plot ln [X] versus time and determine the magnitude of the slope.

(B) Plot 1/[X] versus time and determine the magnitude of the slope.

(C) Run another trial of the experiment with a different initial concentration, plot the data on the same graph, and see where the curves intersect.

(D) Run another trial of the experiment at a different temperature, plot the data on the same graph, and see where the curves have the same slope.

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3. Answer the following questions about ozone

a. The O_3 molecule has a central oxygen atom bonded to two outer oxygen atoms that are not bonded to one another. In the box below, draw the Lewis electron-dot diagram of the O_3 molecule. Include all valid resonance structures.

b. Based on the diagram you drew in part (a), what is the shape of the ozone molecule?

Ozone decomposes according to the reaction represented below

$$2\ O_{3(}g) \rightarrow 3\ O_2(g)$$

$$\Delta H° = -\ 285\ kJ/mol_{rxn}$$

c. The bond enthalpy of the oxygen-oxygen bond in O_2 is 498 kJ/mol. Based on the enthalpy of the reaction represented above, what is the average bond enthalpy, in kJ/mol, of an oxygen-oxygen bond in O_3?

Ozone can oxidize $HSO_3^-(aq)$, as represented by the equation below.

$$HSO_3^-(aq) + O_3(aq) \rightarrow HSO_4^-(aq) + O_2(g)$$

A solution is prepared in which the initial concentration of $HSO_3^-(aq)$ (6.4×10^{-4} M) is much larger than that of $O_3(aq)$ (1.0×10^{-5} M). The concentration of $O_3(aq)$ is monitored as the reaction proceeds, and the data are plotted in the graph below

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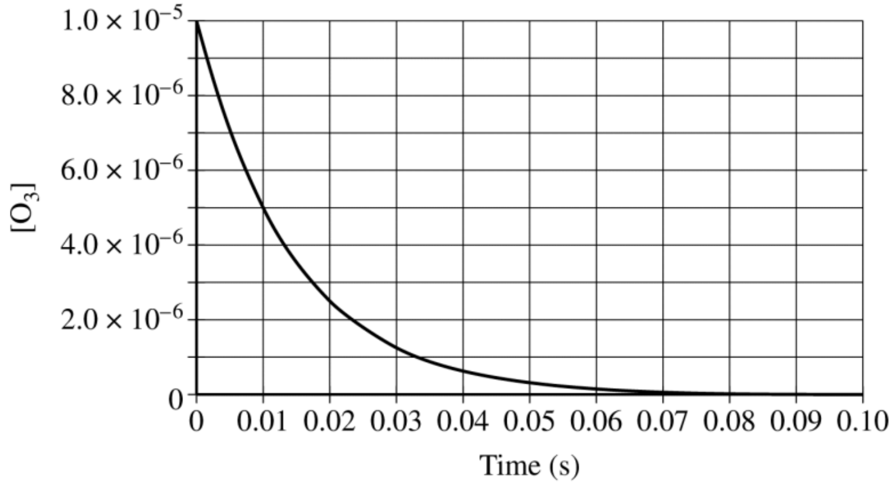

d. The data are consistent with the following rate law: rate = $k_1[O_3]$

i. Based on the graph on the previous page, determine the half-life of the reaction

ii. Determine the value of the rate constant, k_1, for the rate law. Include units with your answer.

iii. Considering the relative concentrations of the reactants, briefly explain why the data in the graph are also consistent with the following rate law: rate = $k_2[O_3][HSO_3^-]$

iv. Briefly describe an experiment that could provide evidence to support the rate law given in part (d)(iii)

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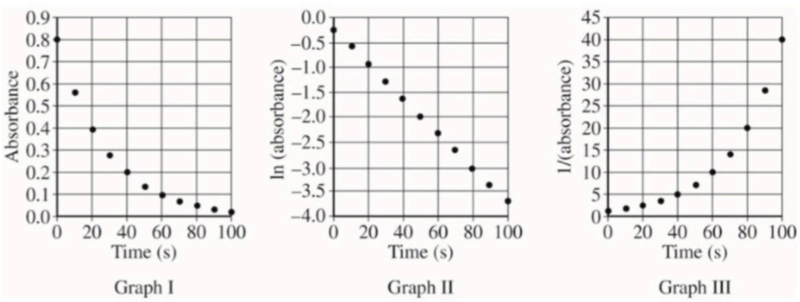
4.

$$\underset{\text{blue}}{Na_2C_{37}H_{34}N_2S_3O_9} + OCl^- \rightarrow \underset{\text{colorless}}{\text{products}}$$

Blue food coloring can be oxidized by household bleach (which contains OCl^-) to form colorless products, as represented by the equation above. A student used a spectrophotometer set at a wavelength of 635 nm to study the absorbance of the food coloring over time during the bleaching process. In the study, bleach is present in large excess so that the concentration of OCl^- is essentially constant throughout the reaction. The student used data from the study to generate the graphs below.



a. Based on the graphs above, what is the order of the reaction with respect to the blue food coloring?

b. The reaction is known to be first order with respect to bleach. In a second experiment, the student prepares solutions of food coloring and bleach with concentrations that differ from those used in the first experiment. When the solutions are combined, the student observes that the reaction mixture reaches an absorbance near zero too rapidly. In order to correct the problem, the student proposes the following three possible modifications to the experiment.

- Increasing the temperature
- Increasing the concentration of the food coloring
- Increasing the concentration of the bleach

Circle the one proposed modification above that could correct the problem, and explain how that modification increases the time for the reaction mixture to reach an absorbance near zero.

c. In another experiment, a student wishes to study the oxidation of red food coloring with bleach. How would the student need to modify the original experimental procedure to determine the order of the reaction with respect to the red food coloring?

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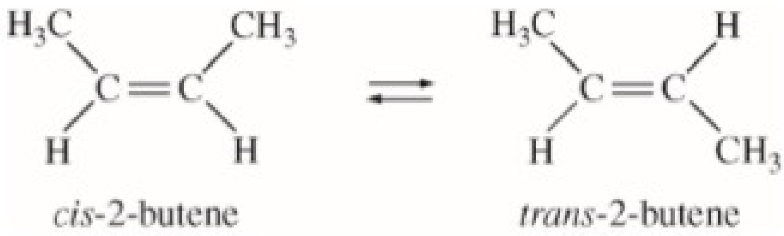
5.



The half-life ($t_{1/2}$) of the catalyzed isomerization of *cis*-2-butene gas to produce *trans*-2-butene gas, represented above, was measured under various conditions, as shown in the table below.

Trial Number	Initial $P_{cis\text{-}2\text{-butene}}$ (torr)	V (L)	T (K)	$t_{1/2}$ (s)
1	300.	2.00	350.	100.
2	600.	2.00	350.	100.
3	300.	4.00	350.	100.
4	300.	2.00	365	50.

a. The reaction is first order. Explain how the data in the table are consistent with a first-order reaction.

b. Calculate the rate constant, k, for the reaction at 350. K. Include appropriate units with your answer.

c. Is the initial rate of the reaction in trial 1 greater than, less than, or equal to the initial rate in trial 2 ? Justify your answer.

d. The half-life of the reaction in trial 4 is less than the half-life in trial 1. Explain why, in terms of activation energy.

6. X → products

Pure substance X decomposes according to the equation above. Which of the following graphs indicates that the rate of decomposition is second order in X ?

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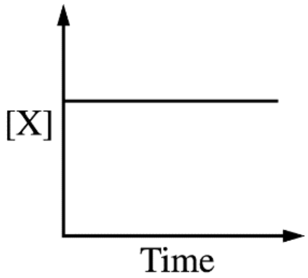
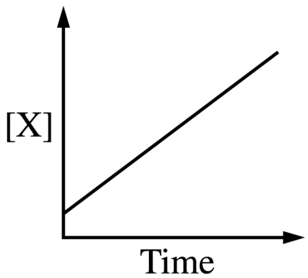
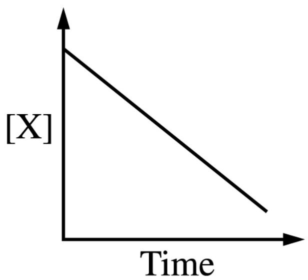
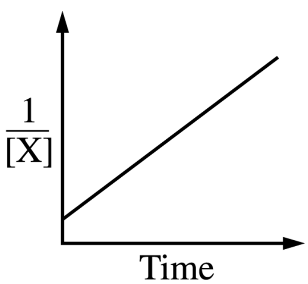
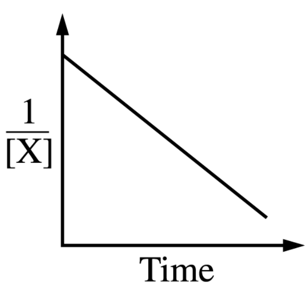
(A) 

(B) 

(C) 

(D) 

(E) 

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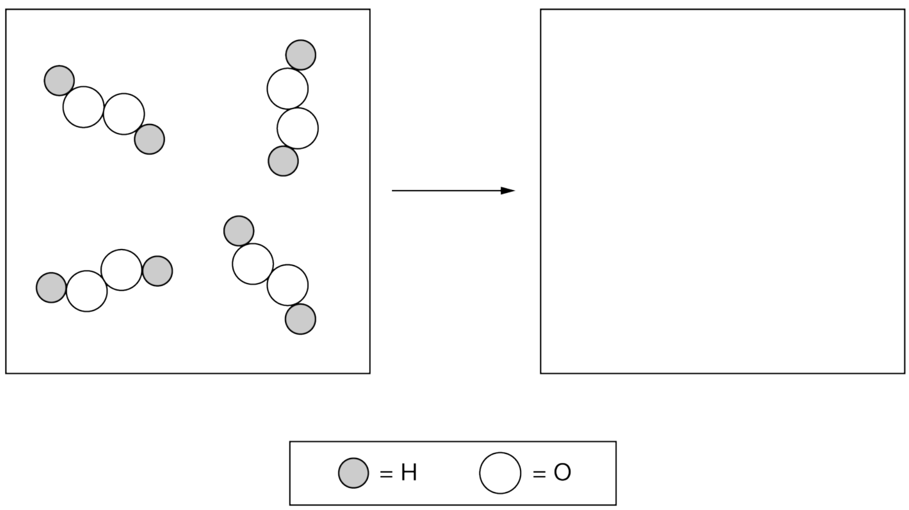
7. **For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

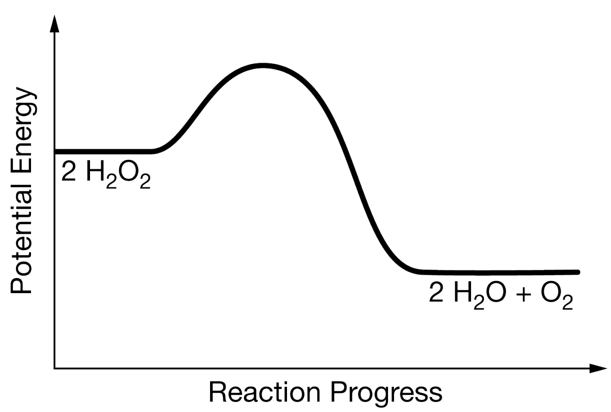
$2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g)$

Hydrogen peroxide, H_2O_2, decomposes according to the equation above. This reaction is thermodynamically favorable at room temperature.

(a) A particulate representation of the reactants is shown below in the box on the left. In the box below on the right, draw the particulate representation of all the molecules that would be produced from these four reactant molecules.



(b) Shown below is a potential energy diagram for the uncatalyzed decomposition of $H_2O_2(aq)$.



(i) According to the diagram, is the decomposition reaction exothermic or endothermic? Justify your answer.

(ii) Manganese dioxide, $MnO_2(s)$, is an insoluble substance that acts as a catalyst for the decomposition

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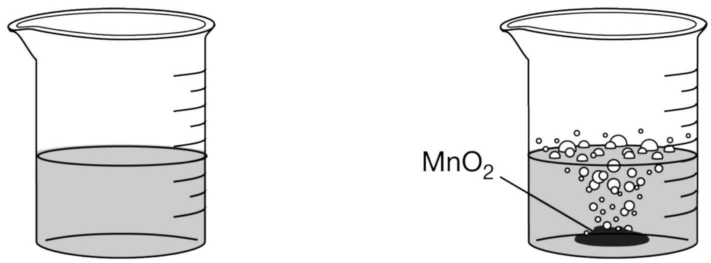
reaction. On the diagram above, draw a curve to represent the reaction as it occurs in the presence of $MnO_2(s)$.

A student investigates the decomposition reaction in the laboratory. The student prepares two small beakers, adding 20.0 mL of $9.77\ M\ H_2O_2(aq)$ to each one. Each beaker is placed on an electronic balance. The student adds 0.10 g of $MnO_2(s)$ to the second beaker and records the mass of each beaker and its contents at 10-second intervals for one minute. The beakers and the data are shown below.



Time (seconds)	Mass of Beaker 1 + $H_2O_2(aq)$ (grams)	Mass of Beaker 2 + $H_2O_2(aq)$ + $MnO_2(s)$ (grams)
0	43.09	43.19
10	43.09	43.06
20	43.09	42.94
30	43.09	42.83
40	43.09	42.73
50	43.09	42.65
60	43.09	42.58

(c) For beaker 2 during the 60-second period, calculate the following.

(i) The number of moles of $O_2(g)$ that was produced

(ii) The mass of $H_2O_2(aq)$ that decomposed

(d) The student continues the experiment for an additional minute. For beaker 2, will the mass of $H_2O_2(aq)$ consumed during the second minute be greater than, less than, or equal to the mass of $H_2O_2(aq)$ consumed during the first minute? Explain your answer referring to the data in the table.

(e) Based on the data, the student claims that the catalyzed reaction has zeroth-order kinetics. Do you agree with the student's claim? Justify your answer.

(f) A second student did the experiment using larger volumes of $9.77\ M\ H_2O_2(aq)$. The student noticed that the reaction in beaker 2 proceeded extremely rapidly, causing some of the liquid to splash out of the beaker onto the lab table. The student claims that as a result of the loss of the liquid from the beaker, the calculated number of

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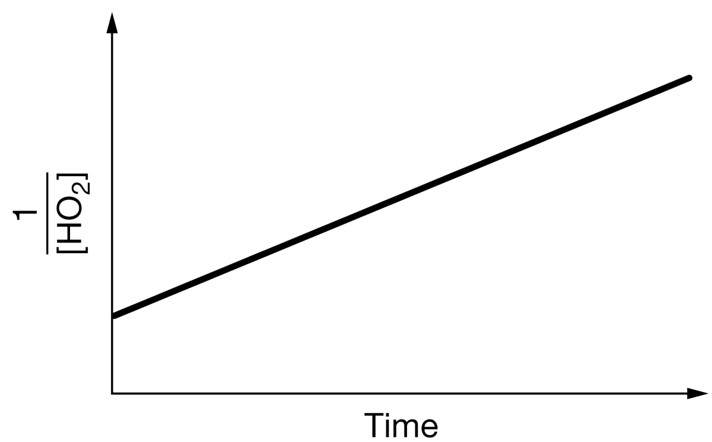
moles of $O_2(g)$ produced is greater than the actual number of moles of $O_2(g)$ produced during the first 60 seconds. Do you agree with the student? Justify your answer.

(g) The hydrogen peroxide used in this experiment can be prepared by the reaction of solid ammonium peroxydisulfate, $(NH_4)_2S_2O_8$, with water. The products are hydrogen peroxide (H_2O_2) and ammonium bisulfate (NH_4HSO_4). Write the balanced equation for the reaction.

8. $2\ HO_2(g) \rightarrow H_2O_2(g) + O_2(g)$

The reaction represented by the chemical equation shown above occurs in Earth's atmosphere. In an experiment, $[HO_2]$ was monitored over time and the data plotted as shown in the following graph.



Based on the information, which of the following is the rate law expression for the reaction?

(A) $\text{Rate} = k$

(B) $\text{Rate} = k[HO_2]$

(C) $\text{Rate} = k[HO_2]^2$

(D) $\text{Rate} = k[H_2O_2][O_2]$

9. After a certain pesticide compound is applied to crops, its decomposition is a first-order reaction with a half-life of 56 days. What is the rate constant, k, for the decomposition reaction?

(A) $0.012\ \text{day}^{-1}$

(B) $0.018\ \text{day}^{-1}$

(C) $56\ \text{day}^{-1}$

(D) $81\ \text{day}^{-1}$

10. After a certain pesticide compound is applied to crops, its decomposition is a first-order reaction with a half-life of 32.5 days. What is the rate constant, k, for the decomposition reaction?

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(A) $0.0213\ \text{day}^{-1}$

(B) $0.0308\ \text{day}^{-1}$

(C) $32.5\ \text{day}^{-1}$

(D) $46.9\ \text{day}^{-1}$

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11. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

An unknown gas, D, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$

A series of experimental trials were performed at $600^\circ\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

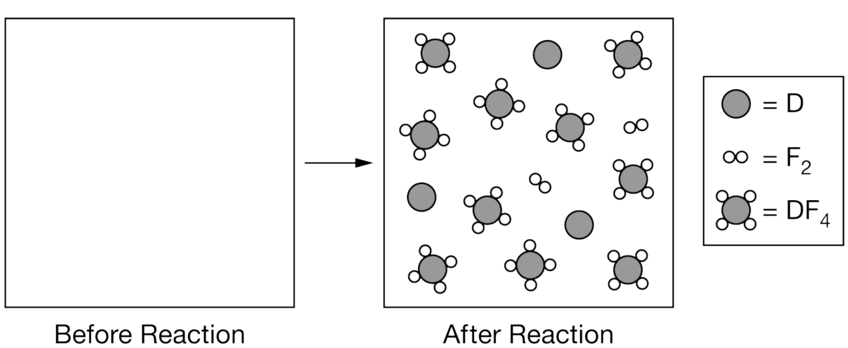
	Initial P_{D} (torr)	Initial $P_{\mathrm{F_2}}$ (torr)	Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min)
Trial 1	1500	1500	5.00×10^{-3}
Trial 2	3000	1500	10.0×10^{-3}
Trial 3	1500	3000	20.0×10^{-3}

(a) A student claims that the reaction is first order with respect to $\mathrm{F_2(g)}$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{F_2}(g)$ in trial 3? Justify your answer.

(c) A student claims that if element D is in group 18, then the molecular geometry of $\mathrm{DF_4}(g)$ is square planar. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to “Before Reaction” is intentionally left blank.)



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(d) If the total pressure at the end of the reaction is $12\ \mathrm{atm}$, what is the partial pressure of $\mathrm{D}(g)$?

(e) Assuming that only D and $\mathrm{F_2}$ are present in the container before the reaction, determine the number of moles of D and $\mathrm{F_2}$ that were present in the initial reaction mixture.

$\mathrm{D} =$ __________

$\mathrm{F_2} =$ __________

The equation and enthalpy change for the reaction between $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ are given below.

$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g) \quad \Delta H^\circ = -250\ \mathrm{kJ/mol}_{rxn}$

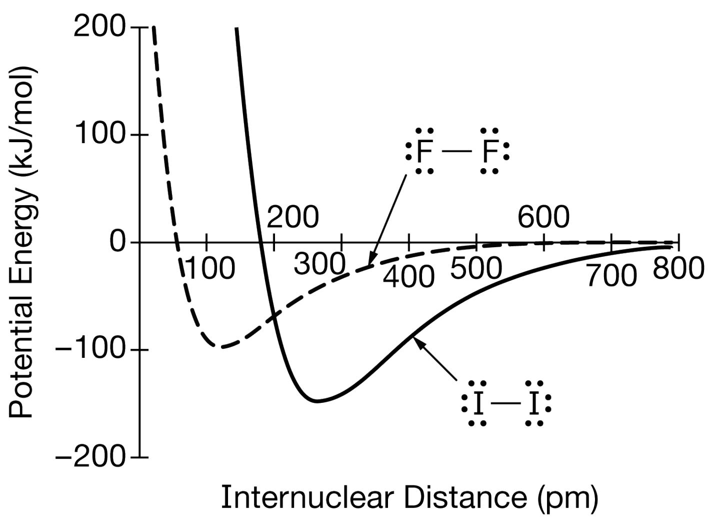
(f) If $0.50\ \mathrm{mol}$ of $\mathrm{DF_4}(g)$ is produced when equimolar amounts of $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ react, how much heat is released?

Fluorine can also react with another gas, $\mathrm{E}(g)$, according to the following equation.

$\mathrm{E}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{EF_4}(g) \quad \Delta \mathrm{H}^\circ = -150\ \mathrm{kJ/mol}_{rxn}$

(g) The student claims that the $\mathrm{D-F}$ bond energy is greater than the $\mathrm{E-F}$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two iodine atoms versus the distance between their nuclei. The student also finds that the $\mathrm{I-I}$ bond energy is slightly less than the $\mathrm{F-F}$ bond energy. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.



(h) Explain the error with the student's sketch for $\mathrm{F_2}(g)$.

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12. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

An unknown gas, D, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$

A series of experimental trials were performed at $600^\circ\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

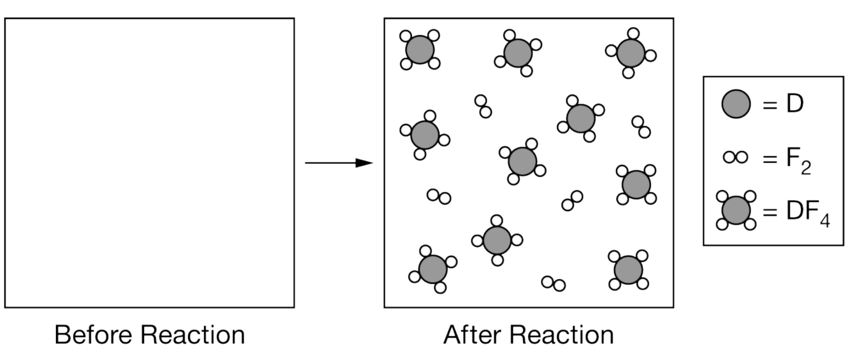
	Initial P_{D} (torr)	Initial $P_{\mathrm{F_2}}$ (torr)	Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min)
Trial 1	3000	3000	2.50×10^{-2}
Trial 2	6000	3000	5.00×10^{-2}
Trial 3	3000	6000	10.0×10^{-2}

(a) A student claims that the reaction is first order with respect to $\mathrm{D}(g)$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{F_2}(g)$ in trial 1? Justify your answer.

(c) A student claims that if element D is in group 18, then the molecular geometry of $\mathrm{DF_4}(g)$ is tetrahedral. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to "Before Reaction" is intentionally left blank.)



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(d) If the total pressure at the end of the reaction is $12\ \mathrm{atm}$, what is the partial pressure of $\mathrm{DF_4}(g)$?

(e) Assuming that only D and $\mathrm{F_2}$ are present in the container before the reaction, determine the number of moles of D and $\mathrm{F_2}$ that were present in the initial reaction mixture.

$\mathrm{D} =$ ___________

$\mathrm{F_2} =$ ___________

The equation and enthalpy change for the reaction between $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ are given below.

$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g) \quad \Delta H^\circ = -250\,\mathrm{kJ/mol}_{rxn}$

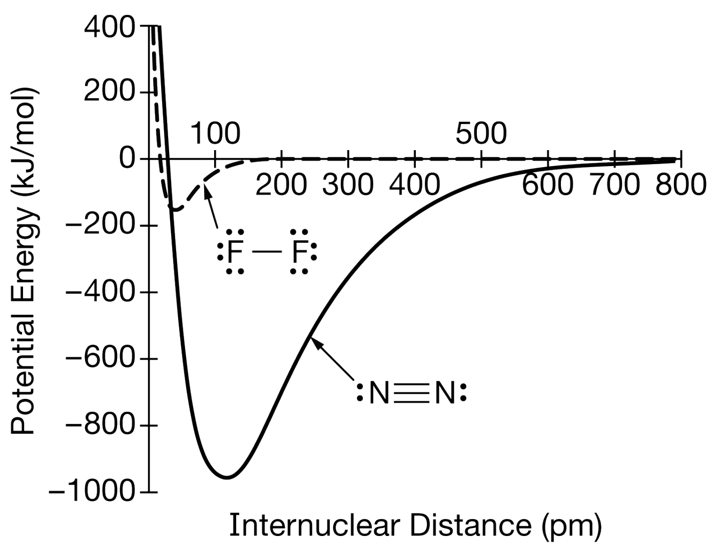
(f) How much heat is released when $4.0\ \mathrm{mol}$ of $\mathrm{F_2}(g)$ reacts with excess $\mathrm{D}(g)$?

Fluorine can also react with another gas, $\mathrm{E}(g)$, according to the following equation.

$\mathrm{E}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{EF_4}(g) \quad \Delta H^\circ = -350\,\mathrm{kJ/mol}_{rxn}$

(g) The student claims that the $\mathrm{D-F}$ bond energy is greater than the $\mathrm{E-F}$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two nitrogen atoms versus the distance between their nuclei. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.



(h) Explain the error with the student's sketch for $\mathrm{F_2}(g)$.

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13. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

An unknown gas, D, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$$

A series of experimental trials were performed at $600^\circ\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

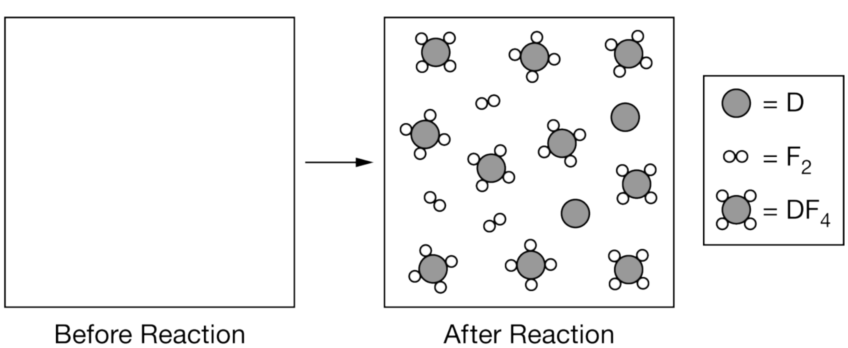
	Initial P_{D} (torr)	Initial $P_{\mathrm{F_2}}$ (torr)	Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min)
Trial 1	3000	3000	2.50×10^{-2}
Trial 2	6000	3000	5.00×10^{-2}
Trial 3	3000	6000	10.0×10^{-2}

(a) A student claims that the reaction is second order with respect to $\mathrm{F_2}(g)$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{D}(g)$ in trial 3? Justify your answer.

(c) A student claims that if element D is in group 16, then the molecular geometry of $\mathrm{DF_4}(g)$ is see-saw. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to "Before Reaction" is intentionally left blank.)



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(d) If the total pressure at the end of the reaction is $12\,\mathrm{atm}$, what is the partial pressure of $\mathrm{D}(g)$?

(e) Assuming that only D and $\mathrm{F_2}$ are present in the container before the reaction, determine the number of moles of D and $\mathrm{F_2}$ that were present in the initial reaction mixture.

$\mathrm{D} =$ __________

$\mathrm{F_2} =$ __________

The equation and enthalpy change for the reaction between $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ are given below.

$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g) \quad \Delta H^\circ = -250\ \mathrm{kJ/mol}_{rxn}$

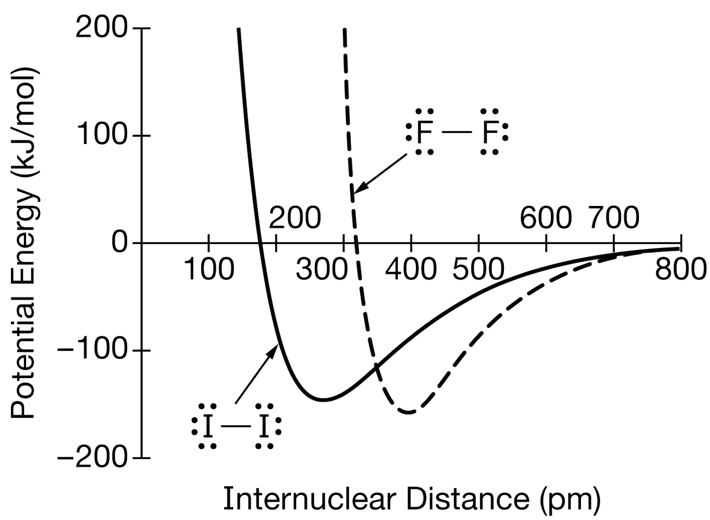
(f) How much heat is released when $0.75\ \mathrm{mol}$ of $\mathrm{D}(g)$ reacts with excess $\mathrm{F_2}(g)$?

Fluorine can also react with another gas, $\mathrm{E}(g)$, according to the following equation.

$\mathrm{E}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{EF_4}(g) \quad \Delta H^\circ = -150\,\mathrm{kJ/mol}_{rxn}$

(g) The student claims that the $\mathrm{D-F}$ bond energy is less than the $\mathrm{E-F}$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the following diagram, which shows the potential energy of two iodine atoms versus the distance between their nuclei. The student also finds that the $\mathrm{I-I}$ bond energy is slightly less than the $\mathrm{F-F}$ bond energy. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.



(h) Explain the error with the student's sketch for $\mathrm{F_2}(g)$.

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14. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

An unknown gas, D, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$

A series of experimental trials were performed at $600^{\circ}\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

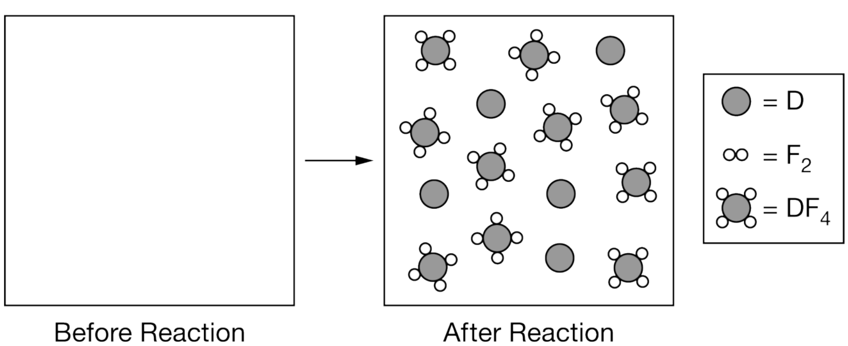
	Initial P_{D} (torr)	Initial $P_{\mathrm{F_2}}$ (torr)	Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min)
Trial 1	3000	3000	2.50×10^{-2}
Trial 2	6000	3000	5.00×10^{-2}
Trial 3	3000	6000	10.0×10^{-2}

(a) A student claims that the reaction is second order with respect to $\mathrm{D}(g)$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{F_2}(g)$ in trial 2? Justify your answer.

(c) A student claims that if element D is in group 16, then the molecular geometry of $\mathrm{DF_4}(g)$ is tetrahedral. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to "Before Reaction" is intentionally left blank.)



Chapter 09 Part 2: Rate Law

(d) If the total pressure at the end of the reaction is $12\,\mathrm{atm}$, what is the partial pressure of $\mathrm{D}(g)$?

(e) Assuming that only D and $\mathrm{F_2}$ are present in the container before the reaction, determine the number of moles of D and $\mathrm{F_2}$ that were present in the initial reaction mixture.

Math output error

The equation and enthalpy change for the reaction between $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ are given below.

$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g) \quad \Delta H^\circ = -250\,\mathrm{kJ/mol}_{rxn}$

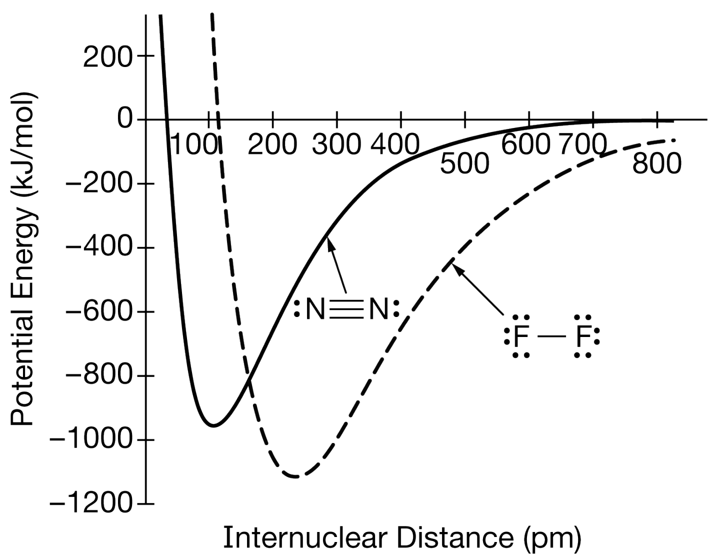
(f) How much heat is released when $0.50\ \mathrm{mol}$ of $\mathrm{F_2}(g)$ reacts with excess $\mathrm{D}(g)$?

Fluorine can also react with another gas, $\mathrm{E}(g)$, according to the following equation.

$\mathrm{E}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{EF_4}(g) \quad \Delta H^\circ = -350\,\mathrm{kJ/mol}_{rxn}$

(g) The student claims that the $\mathrm{D-F}$ bond energy is less than the $\mathrm{E-F}$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two nitrogen atoms versus the distance between their nuclei. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.



(h) Explain the error with the student's sketch for $\mathrm{F_2}(g)$.

15. If the oxygen isotope $^{20}\mathrm{O}$ has a half-life of 15 seconds, what fraction of a sample of pure $^{20}\mathrm{O}$ remains after 1.0 minute?

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{7}{30}$

(D) $\frac{1}{8}$

(E) $\frac{1}{16}$

16. Gaseous cyclobutene undergoes a first-order reaction to form gaseous butadiene. At a particular temperature, the partial pressure of cyclobutene in the reaction vessel drops to one-eighth its original value in 124 seconds. What is the half-life for this reaction at this temperature?

(A) 15.5 sec

(B) 31.0 sec

(C) 41.3 sec

(D) 62.0 sec

(E) 124 sec

17. The isomerization of cyclopropane to propylene is a first-order process with a half-life of 19 minutes at $500^{\circ}C$. The time it takes for the partial pressure of cyclopropane to decrease from 1.0 atmosphere to 0.125 atmosphere at $500^{\circ}C$ is closest to

(A) 38 minutes

(B) 57 minutes

(C) 76 minutes

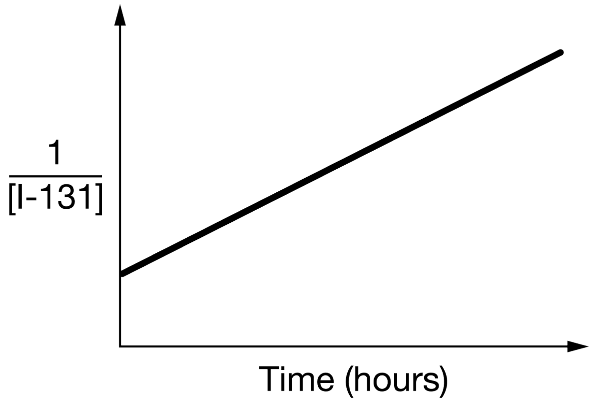
(D) 152 minutes

(E) 190 minutes

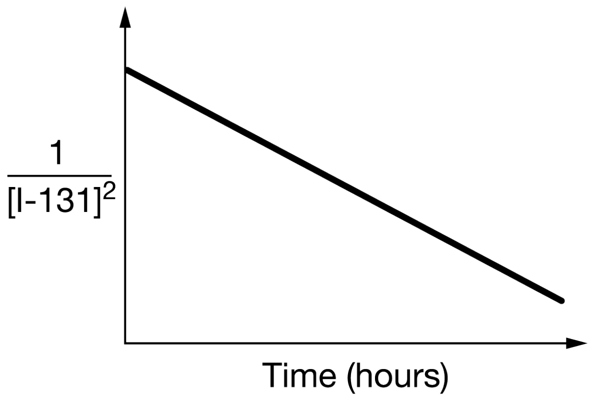
18. The rate constant (k) for the decay of the radioactive isotope I-131 is $3.6 \times 10^{-3}\ \text{hours}^{-1}$. The slope of which of the following graphs is correct for the decay and could be used to confirm the value of k ?

Chapter 09 Part 2: Rate Law

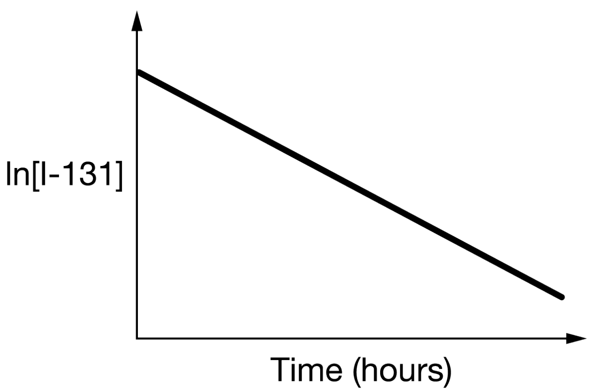
(A)



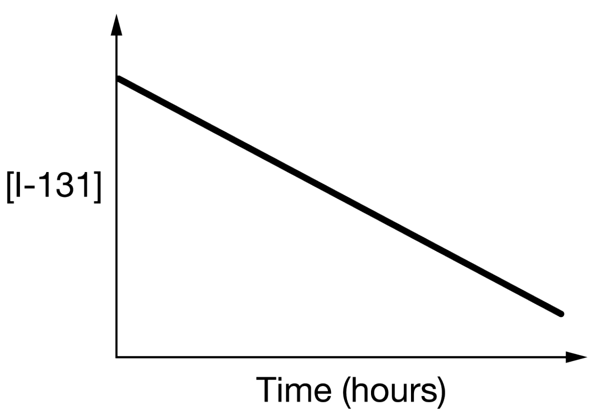
(B)



(C)



(D)



Chapter 09 Part 2: Rate Law

$$2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g) \qquad \Delta H° = -196\ \text{kJ/mol}_{rxn}$$

The decomposition of $H_2O_2(aq)$ is represented by the equation above. A student monitored the decomposition of a 1.0 L sample of $H_2O_2(aq)$ at a constant temperature of 300. K and recorded the concentration of H_2O_2 as a function of time. The results are given in the table below.

Time (s)	$[H_2O_2]$
0	2.7
200.	2.1
400.	1.7
600.	1.4

19. Which of the following statements is a correct interpretation of the data regarding how the order of the reaction can be determined?

(A) The reaction must be first order because there is only one reactant species.

(B) The reaction is first order if the plot of ln $[H_2O_2]$ versus time is a straight line.

(C) The reaction is first order if the plot of $1/[H_2O_2]$ versus time is a straight line.

(D) The reaction is second order because 2 is the coefficient of H_2O_2 in the chemical equation.

20.

$$H_3AsO_4 + 3\ I^- + 2\ H3O^+ \rightarrow H_3AsO_3 + I_3^- + H_2O$$

The oxidation of iodide ions by arsenic acid in acidic aqueous solution occurs according to the stoichiometry shown above. The experimental rate law of the reaction is:

$$\text{Rate} = k[H_3AsO_4]\ [I^-]\ [H_3O^+]$$

What is the order of the reaction with respect to I^-?

(A) 1

(B) 2

(C) 3

(D) 5

(E) 6

Chapter 09 Part 2: Rate Law

21. If 87.5 percent of a sample of pure ^{131}I decays in 24 days, what is the half-life of ^{131}I?

(A) 6 days
(B) 8 days
(C) 12 days
(D) 14 days
(E) 21 days

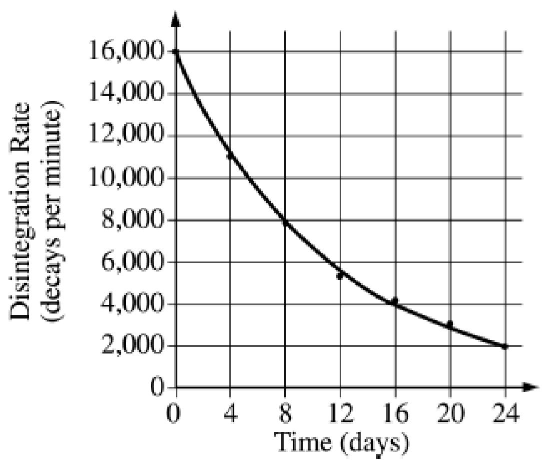
Chapter 09 Part 2: Rate Law

22. The decay of the radioisotope I-131 was studied in a laboratory. I-131 is known to decay by beta $\left({}^{0}_{-1}e\right)$emission.

a. Write a balanced nuclear equation for the decay of I-131.

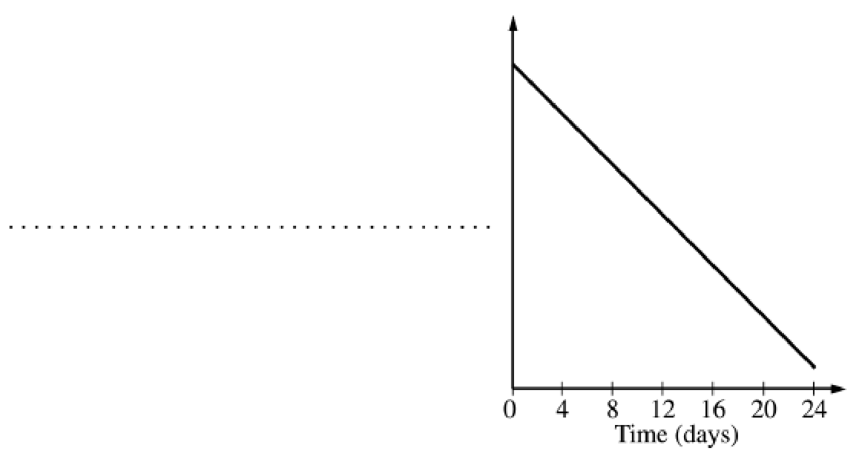
b. What is the source of the beta particle emitted from the nucleus?

The radioactivity of a sample of I-131 was measured. The data collected are plotted on the graph below.



c. Determine the half-life, $t_{1/2}$, of I-131 using the graph above.

d. The data can be used to show that the decay of I-131 is a first-order reaction, as indicated on the graph below.



i. Label the vertical axis of the graph above.

ii. What are the units of the rate constant, k, for the decay reaction?

Chapter 09 Part 2: Rate Law

iii. Explain how the half-life of I-131 can be calculated using the slope of the line plotted on the graph.

e. Compare the value of the half-life of I-131 at 25°C to its value at 50°C.

23. The half-life of ^{55}Cr is about 2.0 hours. The delivery of a sample of this isotope from the reactor to a certain laboratory requires 12 hours. About what mass of such material should be shipped in order that 1.0 mg of ^{55}Cr is delivered to the laboratory?

(A) 130 mg

(B) 64 mg

(C) 32 mg

(D) 11 mg

(E) 1.0 mg

24.

Experiment	Initial [NO] (mol L^{-1})	Initial [O_2] (mol L^{-1})	Initial Rate of Formation of NO_2 (mol $L^{-1}s^{-1}$)
1	0.10	0.10	2.5×10^{-4}
2	0.20	0.10	5.0×10^{-4}
3	0.20	0.40	8.0×10^{-3}

The initial-rate data in the table above were obtained for the reaction represented below. What is the experimental rate law for the reaction?

$$2\,NO(g) + O_2(g) \rightarrow NO_2(g)$$

(A) Rate = $k[NO][O_2]$

(B) Rate = $k[NO][O_2]^2$

(C) Rate = $k[NO]^2[O_2]$

(D) Rate = $k[NO]^2[O_2]^2$

(E) Rate = $k[NO] / [O_2]$

25.

Experiment	$[X]_0$	$[Y]_0$	Initial Rate of Formation of Z (mol L^{-1} sec^{-1})
1	0.40	0.10	R
2	0.20	0.20	?

The table above shows the results from a rate study of the reaction X + Y → Z. Starting with known concentrations of X and Y in experiment 1, the rate of formation of Z was measured. If the reaction was first order with respect to X and second order with respect to Y, the initial rate of formation of Z in experiment 2 would be

(A) $\frac{R}{4}$

(B) $\frac{R}{2}$

(C) R

(D) $2R$

(E) $4R$

26.

$$\text{Rate} = k[\text{M}][\text{N}]^2$$

The rate of a certain chemical reaction between substances M and N obeys the rate law above. The reaction is first studied with [M] and [N] each 1 x 10^{-3} molar. If a new experiment is conducted with [M] and [N] each 2 x 10^{-3} molar, the reaction rate will increase by a factor of

(A) 2

(B) 4

(C) 6

(D) 8

(E) 16

27.

Trial	$[\text{A}_2]$	$[\text{B}]$	**Initial rate** $(M\text{s}^{-1})$
1	0.10	0.50	2.5×10^{-4}
2	0.20	0.50	5.0×10^{-4}
3	0.30	0.05	5.0×10^{-5}
4	0.30	0.10	1.0×10^{-4}

An experiment was conducted to determine the rate law for the reaction $\text{A}_2(g) + \text{B}(g) \rightarrow \text{A}_2\text{B}(g)$. The table above shows the data collected. Based on the data in the table, which statement is correct?

(A) Since the rate law can be expressed as $rate = k[\text{A}_2]^2$, tripling the concentration of A_2 will cause a 9-fold increase in the rate of the reaction.

(B) Since the rate law can be expressed as $rate = k[\text{A}_2][\text{B}]$, doubling the concentrations of A_2 and B will quadruple the rate of the reaction.

(C) Since the rate law can be expressed as $rate = k[\text{A}_2]^2[\text{B}]$, tripling the concentration of A_2 while keeping the concentration of B constant will triple the rate of the reaction.

(D) Since the rate law can be expressed as $rate = k[\text{A}_2][\text{B}]^2$, doubling the concentration of B while keeping the concentration of A_2 constant will double the rate of the reaction.

Chapter 09 Part 2: Rate Law

28.

Time (s)	$[NO_2]$ (mol/L)	ln $[NO_2]$	$\frac{1}{[NO_2]}$ (L/mol)
0	0.500	−0.693	2.00
100	0.364	−1.01	2.75
200	0.286	−1.25	3.50
300	0.235	−1.45	4.25

The data from a study of the decomposition of $NO_2(g)$ to form $NO(g)$ and $O_2(g)$ are given in the table above. Which of the following rate laws is consistent with the data?

(A) $\text{Rate} = k[NO_2]$

(B) $\text{Rate} = k[NO_2]^2$

(C) $\text{Rate} = k\frac{1}{[NO_2]}$

(D) $\text{Rate} = k\frac{1}{[NO_2]^2}$

29. $2\ NO(g) + Cl_2(g) \rightarrow 2\ NOCl(g)$

Experiment	[NO]	$[Cl_2]$	**Initial Rate of** $NOCl$ **Formation** $(M\ s^{-1})$
1	0.0250	0.0510	4.55×10^{-5}
2	0.0250	0.1020	9.10×10^{-5}
3	0.0500	0.0510	1.82×10^{-4}

The initial rates of the reaction represented by the equation shown above were measured for different initial concentrations of $NO(g)$ and $Cl_2(g)$. Based on the data given in the table above, which of the following is the rate law expression for the reaction, and why?

(A) $\text{Rate} = k[NO]^2$, because the initial rate quadrupled when $[NO]$ was doubled but remained constant when $[Cl_2]$ was doubled.

(B) $\text{Rate} = k[NO][Cl_2]$, because the initial rate doubled when either $[NO]$ or $[Cl_2]$ was doubled.

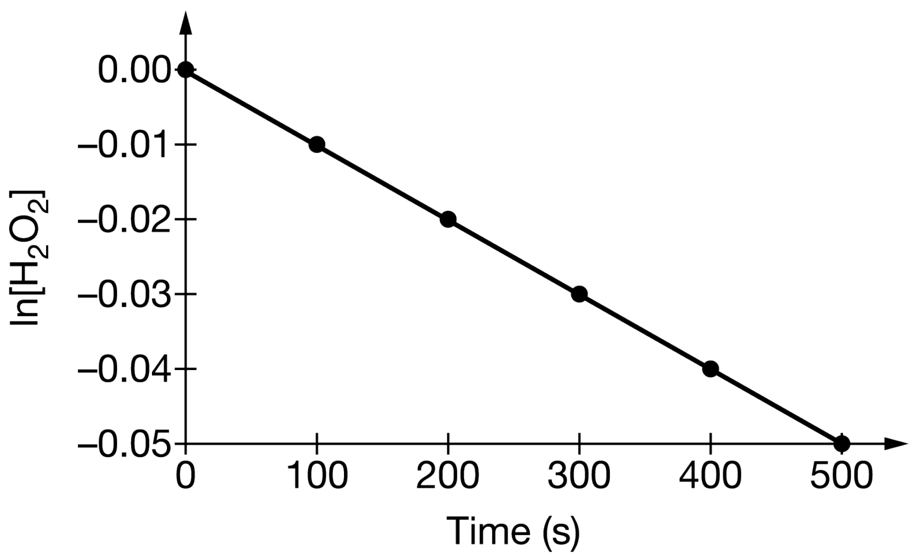
(C) $\text{Rate} = k[NO][Cl_2]^2$, because the initial rate doubled when $[NO]$ was doubled and quadrupled when $[Cl_2]$ was doubled.

(D) $\text{Rate} = k[NO]^2[Cl_2]$, because the initial rate quadrupled when $[NO]$ was doubled and doubled when $[Cl_2]$ was doubled.

Chapter 09 Part 2: Rate Law

30. $H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$

Hydrogen peroxide decomposes to produce water and oxygen according to the equation above. An experimentally determined graph for the first-order decomposition of hydrogen peroxide is provided below.



Which of the following best identifies the rate constant k for the reaction based on the information in the plot of $\ln[H_2O_2]$ versus time (t) ?

(A) $k = \ln[H_2O_2]$ at $t = 0$ s

(B) $k = \ln[H_2O_2]$ at $t = 500$ s

(C) $k = -(\text{slope of plot})$

(D) $k = -\frac{1}{(\text{slope of plot})}$

31. $S_2O_8{}^{2-}(aq) + 3\,I^-(aq) \rightarrow 2\,SO_4{}^{2-}(aq) + I_3{}^-(aq)$

In aqueous solution, the reaction represented by the balanced equation shown above has the experimentally determined rate law: $\text{rate} = k[S_2O_8{}^{2-}][I^-]$.

If the concentration of $[S_2O_8{}^{2-}]$ is doubled while keeping $[I^-]$ constant, which of the following experimental results is predicted based on the rate law, and why?

(A) The rate of reaction will remain the same, because k will decrease by half.

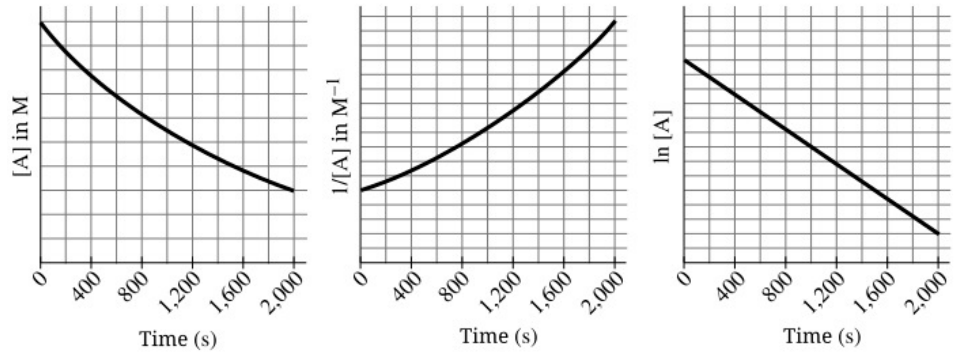
(B) The rate of reaction will double, because the rate is directly proportional to $[S_2O_8{}^{2-}]$.

(C) The rate of reaction will increase by a factor of four, because two moles of $SO_4{}^{2-}$ are produced for each mole of $S_2O_8{}^{2-}$ consumed.

(D) The rate of reaction will increase by a factor of four, because the reaction is second order overall.

Chapter 09 Part 2: Rate Law

32.



A student recorded the rate of decomposition of compound A as a function of time at $45^\circ\mathrm{C}$. The experimental data are shown above. What is the rate law for the decomposition of A at this temperature?

(A) Rate $= k$

(B) Rate $= k[\mathrm{A}]$

(C) Rate $= k[\mathrm{A}]^2$

(D) The rate law cannot be determined without additional experimentation.

33. The rate law for the reaction of nitrogen dioxide and chlorine is found to be rate = $k\,[NO_2]^2[Cl_2]$. By what factor does the rate of the reaction change when the concentrations of both NO_2 and Cl_2 are doubled?

(A) 2

(B) 3

(C) 4

(D) 6

(E) 8

34.

Experiment	Initial [X] (mol L^{-1})	Initial [Y] (mol L^{-1})	Initial Rate of Formation of Z (mol L^{-1} s^{-1})
1	0.10	0.30	4.0×10^{-4}
2	0.20	0.60	1.6×10^{-3}
3	0.20	0.30	4.0×10^{-4}

The data in the table above were obtained for the reaction $X + Y \rightarrow Z$. Which of the following is the rate law for the reaction?

(A) Rate = $k[X]^2$

(B) Rate = $k[Y]^2$

(C) Rate = $k[X][Y]$

(D) Rate = $k[X]^2[Y]$

(E) Rate = $k[X][Y]^2$

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35.

Time (days)	0	1	2	3	4	5	6	7	...	10	...	20
% Reactant remaining	100	79	63	50	40	31	25	20		10		1

A reaction was observed for 20 days and the percentage of the reactant remaining after each day was recorded in the table above. Which of the following best describes the order and the half-life of the reaction?

(A)

Reaction Order	Half-life (days)
First	3

(B)

Reaction Order	Half-life (days)
First	10

(C)

Reaction Order	Half-life (days)
Second	3

(D)

Reaction Order	Half-life (days)
Second	3

(E)

Reaction Order	Half-life (days)
Second	10

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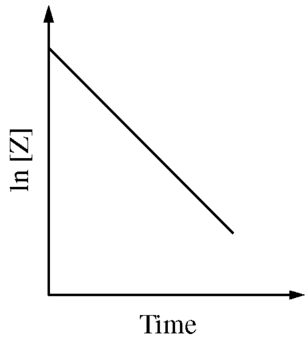
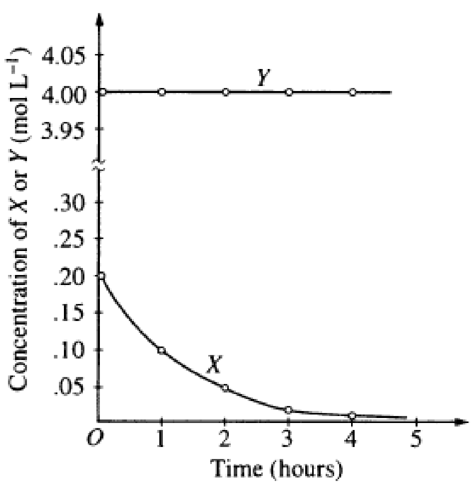
36.



Consider the reaction represented by the equation 2 X + 2 Z → X_2Z_2. During a reaction in which a large excess of reactant X was present, the concentration of reactant Z was monitored over time. A plot of the natural logarithm of the concentration of Z versus time is shown in the figure above. The order of the reaction with respect to reactant Z is

(A) zero order
(B) first order
(C) second order
(D) third order

37.



The graph above shows the results of a study of the reaction of *X* with a large excess of *Y* to yield *Z*. The concentrations of *X* and *Y* were measured over a period of time. According to the results, which of the following can be concluded about the rate law for the reaction under the conditions studied?

(A) It is zero order in [*X*].
(B) It is first order in [*X*].
(C) It is second order in [*X*].
(D) It is first order in [*Y*].
(E) The overall order of the reaction is 2.

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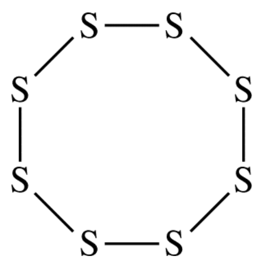
38.



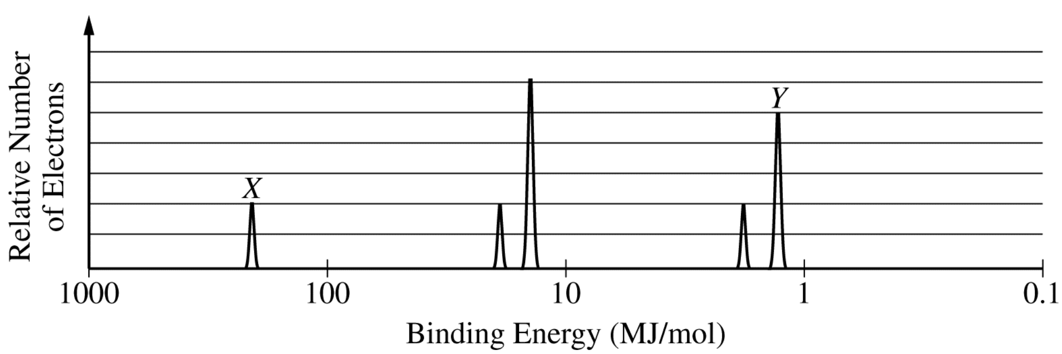
Elemental sulfur can exist as molecules with the formula S_8. The S_8 molecule is represented by the incomplete Lewis diagram above.

a. The diagram of S_8 shows only bonding pairs of electrons. How many lone pairs of electrons does each S atom in the molecule have?

b. Based on your answer to part (a), determine the expected value of the S–S–S bond angles in the S_8 molecule.

c. Write the electron configuration for the S atom in its ground state.

d. The complete photoelectron spectrum for the element chlorine is represented below. Peak *X* in the spectrum corresponds to the binding energy of electrons in a certain orbital of chlorine atoms. The electrons in this orbital of chlorine have a binding energy of 273 MJ/mol, while the electrons in the same orbital of sulfur atoms have a binding energy of 239 MJ/mol



i. Identify the orbital and explain the difference between the binding energies in terms of Coulombic forces.

ii. Peak *Y* corresponds to the electrons in certain orbitals of chlorine atoms. On the spectrum shown, carefully draw the peak that would correspond to the electrons in the same orbitals of sulfur atoms.

$3S_8 + 8OH^- \rightarrow 8S_3^- + 4HOOH$

In an experiment, a student studies the kinetics of the reaction represented above and obtains the data shown in the following table.

Experiment	Initial $[S_8]$ (M)	Initial $[OH^-]$ (M)	Initial Reaction Rate (M/s)
Trial 1	0.100	0.0100	0.699
Trial 2	0.300	0.0100	2.10
Trial 3	0.300	?	4.19

e. Use the data in the table to do the following.

i. Determine the order of the reaction with respect to S_8. Justify your answer.

ii. Determine the value of $[OH^-]$ that was used in trial 3, considering that the reaction is first order with respect to OH^- . Justify your answer.

The next day the student conducts trial 4 using the same concentrations of S_8 and OH^- as in trial 1, but the reaction occurs at a much slower rate than the reaction in trial 1. The student observes that the temperature in the lab is lower than it was the day before.

f. Using particle-level reasoning, provide TWO explanations that help to account for the fact that the reaction rate is slower in trial 4.

39.

$$H_3AsO_4 + 3\ I^- + 2\ H3O^+ \rightarrow H_3AsO_3 + I_3^- + H_2O$$

The oxidation of iodide ions by arsenic acid in acidic aqueous solution occurs according to the stoichiometry shown above. The experimental rate law of the reaction is:

$$\text{Rate} = k[H_3AsO_4]\ [I^-]\ [H_3O^+]$$

According to the rate law for the reaction, an increase in the concentration of hydronium ion has what effect on this reaction?

(A) The rate of reaction increases.
(B) The rate of reaction decreases.
(C) The value of the equilibrium constant increases.
(D) The value of the equilibrium constant decreases.
(E) Neither the rate nor the value of the equilibrium constant is changed.

40. $2\ N_2O(g) \rightarrow 2\ N_2(g) + O_2(g)$

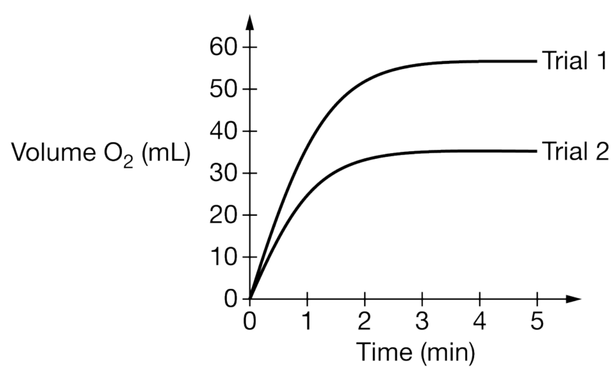
The rate law for the reaction represented above is experimentally determined to be $\text{rate} = k[N_2O]$. Which of the following best explains why increasing the initial concentration of $N_2O(g)$ at a constant temperature increases the reaction rate?

(A) Molecular collisions become more frequent.
(B) There is a higher probability of effective molecular collisions.
(C) The rate constant for the reaction increases.
(D) The reaction mechanism changes to one that has a lower activation energy.

41. $2\ H_2O_2(aq) \xrightarrow{KI(s)} 2\ H_2O(l) + O_2(g)$

$\text{Rate} = k[H_2O_2]$

A student performed an experiment to study the factors that affect the rate of the first-order catalytic decomposition of $H_2O_2(aq)$, represented above.



The student monitored the volume of $O_2(g)$ produced over time as the reaction proceeded. The data from two trials are plotted in the graph above. Which of the following best explains the results of trial 2 compared with those of trial 1 ?

(A) The reaction temperature in trial 2 was higher than it was in trial 1.
(B) The volume of the reaction vessel was larger in trial 2 than it was in trial 1.
(C) The student used a smaller amount of $KI(s)$ in trial 2 compared with that used in trial 1.
(D) The concentration of $H_2O_2(aq)$ was lower in trial 2 than it was in trial 1.